

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. Light travels fastest in vacuum. Suppose light takes 2.0 seconds to travel from the Sun to Earth through space (vacuum). If the same distance were travelled entirely through water, and light is $(3/4)$ times slower in water than in vacuum, what would be the new time taken?
(A) 1.5 s (B) 2.0 s
(C) 2.67 s (D) 3.5 s

2. Fill in the blanks from the given table by selecting an appropriate option.

S. No	Substance	Colour change with phenolphthalein indicator	Colour changes with Methyl orange indicator
1	Grape juice	Colourless	(iii)
2	Sea water	(i)	Yellow
3	Aloe vera gel	(ii)	Yellow
4	Pineapple	Colourless	(iv)

	(i)	(ii)	(iii)	(iv)
(A)	Pink	Light pink	Red	Red
(B)	Colourless	Light pink	Colourless	Red
(C)	Red	Colourless	Pink	Light pink
(D)	Light pink	Red	Colourless	Colourless

3. A patient is found to have a heart valve that does not close completely (a condition called "Regurgitation"). During the contraction of the Left Ventricle, some blood flows backward into the Left Atrium. **What would be the most immediate physiological effect on this patient?**
(A) The blood pressure in the arteries will increase significantly.
(B) The body tissues will receive less oxygenated blood per heartbeat, leading to fatigue.
(C) The lungs will stop producing Carbon dioxide.
(D) The heart rate will drop to 30 beats per minute to compensate.

4. Some Examples of changes are given below:

- Crushing of can
- Dissolving sugar in water
- Chopping wood
- Boiling water
- Heating a metal rod

Classify the above changes into

I. Irreversible physical change

II. Reversible physical change

(A) I-a,c II-b,d,e

(B) I-b,d II-a,c,e

(C) I-d,a,e II-b,c

(D) I-e,c,a II-b,d

5. **Assertion (A):** Vegetative propagation is a faster method of reproduction compared to sexual reproduction from seeds.

Reason (R): Plants produced by vegetative propagation are genetically different from the parent plant, allowing them to adapt faster.

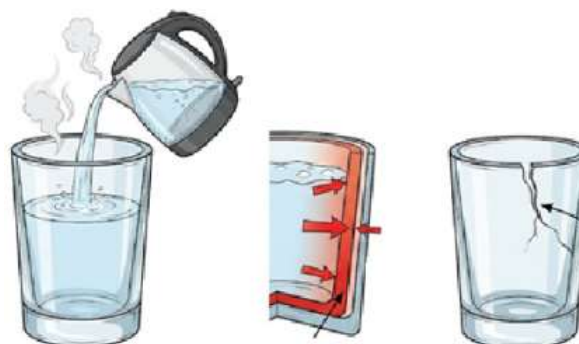
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

6. A thick glass tumbler cracks when boiling water is suddenly poured into it. Which of the following best explains the mechanism of this fracture?



(A) The inner surface expands faster than the outer surface, causing strain.

(B) The outer surface expands, compressing the inner surface.

(C) Glass is a good conductor, so heat leaves too quickly.

(D) The boiling water reacts chemically with the glass surface.

7. A city is rapidly expanding, replacing open dirt fields and parks with concrete roads and multi-story apartment buildings.

What is the most likely consequence of this expansion on the local water table?

(A) The water table will rise due to less transpiration.

(B) The water table will drop because water cannot seep into the ground (infiltration).

(C) The water table will remain the same.

(D) Rainwater will purify itself on the concrete before reaching the ground.

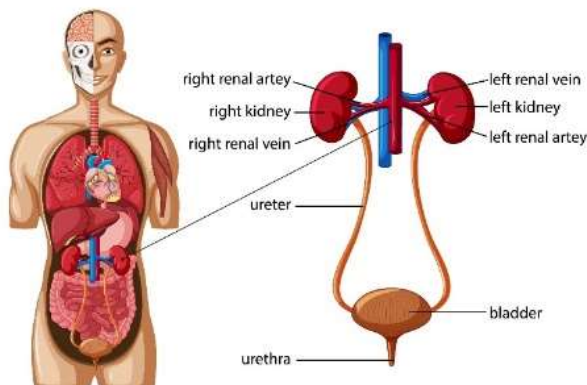
8. Sai took a turmeric paper and tried to draw a tree on it with the help of cotton sticks dipped in solutions P and Q. The tree drawn with solution P was not visible while one drawn with solution Q

turned red Solution P and Q could be respectively

- (A) Lime water and lemon juice
- (B) Vinegar and soap solution
- (C) Common salt solution and baking soda
- (D) Both B and C

9. The diagram below shows the human excretory system. What would happen if the tubes labeled 'Ureters' were blocked by a kidney stone?

Human Kidneys and Bladder

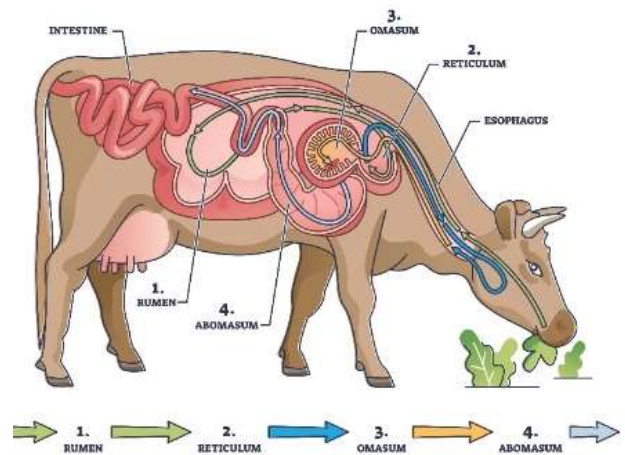


- (A) The kidneys would stop producing urine immediately.
- (B) Urine would be unable to travel from the kidneys to the urinary bladder, causing swelling and pain in the kidneys.
- (C) The blood would stop being filtered in the Bowman's capsule.
- (D) Urine would flow directly into the large intestine instead.

10. A cow quickly swallows grass and stores it in a specific part of its stomach called the rumen. Later, the food returns to the mouth in small lumps for the cow to chew. Which of the following best

describes the biological necessity of this two-step process?

RUMINANT DIGESTION



- (A) The grass contains high levels of starch that require double saliva mixing.
 - (B) The rumen contains symbiotic bacteria that break down cellulose, which the cow cannot digest on its own.
 - (C) The process is meant to cool down the food before it enters the small intestine.
 - (D) Chewing twice helps in the absorption of water directly through the tongue.
11. A person is suffering from a condition where their "Small Intestine" has a very smooth inner lining with almost no villi. Which of the following is the most likely symptom of this condition?
- (A) Severe constipation due to too much water absorption.
 - (B) Malnutrition and weight loss, even if the person eats a large amount of food.
 - (C) Inability to breathe properly due to lack of glucose in the lungs.
 - (D) Rapid increase in the production of saliva in the mouth.

12. Iron filings are mixed with sulfur powder. The mixture is heated strongly to form iron sulfide. Which observation best proves that a chemical change has occurred?

- (A) The mixture becomes hot
- (B) The color of the substance changes
- (C) The original substances cannot be separated by a magnet
- (D) The mass of the system remains the same

13. In a medical simulation, a patient's kidneys have stopped functioning. They are put on a dialysis machine. Which physical principle is the dialysis machine mimicking to clean the blood?

- (A) Active transport of urea against a concentration gradient using ATP.
- (B) Diffusion of waste products across a semi-permeable membrane into a dialyzing solution.
- (C) Filtration of blood through high-pressure pumps located in the bladder.
- (D) Chemical neutralization of toxins using alkaline substances in the blood.

14. A student argues that because 71% of the earth's surface is covered by water, water scarcity is a myth. What is the critical counter-argument based on available data?

- (A) 97% of the earth's water is in the ocean, which is too saline for drinking or farming.
- (B) Only 0.5% of the earth's water is groundwater.
- (C) Most freshwater is in the atmosphere.
- (D) The oceans are shrinking.

15. The table represents the **electrical conductivity** of four solutions:

Solution	Conductivity
HCl	High
CH ₃ COOH	Low
NaOH	High
NaCl	Medium

Which conclusion is **most valid**?

- (A) All acids conduct electricity equally
- (B) Strong acids and bases ionize completely
- (C) Salts do not conduct electricity
- (D) Weak acids are neutral

16. The data given below represents the amount of heat released during four reactions.

- Reaction A – 0 kJ
- Reaction B – +20 kJ
- Reaction C – –15 kJ
- Reaction D – 25 kJ

Which reaction is endothermic?

- (A) A
- (B) B
- (C) C
- (D) D

17. A town is located in a semi-arid region where rainfall is scarce and irregular. Over the past decade, groundwater levels have been dropping steadily due to over-extraction for agriculture and domestic use. The local government is considering constructing a large dam on a nearby river to provide water for irrigation and drinking purposes.

Which of the following statements provides the most balanced view regarding this solution?

(A) Building a dam will completely solve the town's water scarcity without any negative consequences.

(B) Constructing a dam can help store water temporarily, but long-term sustainability requires efficient water use, rainwater harvesting, and reducing wastage.

(C) Dams are useless because they evaporate water faster than it is replenished.

(D) The town should rely entirely on groundwater and avoid using river water.

18. A thermometer scale has the lower fixed point marked at **10°C** instead of **0°C** and upper fixed point at **110°C** instead of **100°C**. If it shows **60°C** on this faulty scale, the correct temperature is:

(A) 40°C (B) 50°C

(C) 60°C (D) 70°C

19. A bread mould (*Rhizopus*) spreads rapidly on a damp slice of bread kept in a dark cupboard. If you observe it under a microscope, you see tiny "blob-on-a-stick" structures. What are these blobs, and how do they function?

(A) They are seeds that were already present in the flour.

(B) They are sporangia containing spores that are released into the air to start new colonies.

(C) They are roots that suck moisture directly from the air.

(D) They are male and female gametes waiting to fuse.

20. A teacher gives a mixture of 25 mL H_2SO_4 (0.2 M) and 25 mL NaOH (0.2 M). After reaction, the resulting solution is divided into three beakers:

- Beaker 1: Add phenolphthalein → solution is colorless
- Beaker 2: Add blue litmus → turns red
- Beaker 3: Add red litmus → remains red

Which explanation is logically correct?

(A) Solution in Beaker 1 is neutral; Na_2SO_4 forms.

(B) Solution in Beaker 2 turns red because of leftover acid.

(C) Beaker 3 remains red because litmus is insensitive to neutral solutions.

(D) Beaker 1 is basic, Beaker 2 is acidic, Beaker 3 is neutral.

21. A metal ring is heated. What happens to the size of the **hole** in the center?

(A) It decreases because the metal expands inwards.

(B) It increases because the metal expands outwards.

(C) It remains the same size.

(D) It first decreases then increases.

22. An experimenter removes the "Anthers" from a flower before they mature but leaves the "Stigma" intact. They then wrap the flower in a fine muslin bag. What is the most likely reason for this experimental procedure?

(A) To prevent the flower from breathing.

(B) To ensure the flower performs self-pollination.

(C) To prevent unwanted self-pollination and control the cross-pollination process.

(D) To kill the flower so it can be studied as a specimen.

23. Assertion (A): Ice cubes floating in water is a physical change.

Reason (R): Ice is less dense than water.

(A) Both A and R are true, and R is the correct explanation of A

(B) Both A and R are true, but R is not the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

24. During an experiment, a student measured their breathing rate. At rest, it was 15 breaths per minute. After running for 10 minutes, it increased to 25 breaths per minute. Why does the body increase the breathing rate during physical activity?

(A) To release the extra heat generated by the friction of the muscles.

(B) To supply more oxygen to the cells for rapid breakdown of food to release more energy.

(C) To prevent the lungs from collapsing under the pressure of running.

(D) To help the heart beat slower by providing more air.

25. A fruit grower notices that his "Seedless Watermelons" do not produce any pollen. How does he manage to grow more seedless watermelons for the next year?

(A) He waits for the watermelon to turn into a seed on its own.

(B) He uses vegetative propagation techniques like "Grafting" or "Tissue Culture" to produce clones of the parent plant.

(C) He prays for a miracle every spring.

(D) He buys regular watermelon seeds and paints them to look seedless.

26. Match the objects in Column I with their type of motion in Column II.

Column I	Column II
(P) Motion of a plucked guitar string	(i) Rectilinear Motion
(Q) Motion of Earth around the Sun	(ii) Oscillatory/Vibratory Motion
(R) A coin moving on a carrom board	(iii) Curvilinear + Rotatory
(S) Motion of a drill bit drilling a hole	(iv) Combined Translatory + Rotatory

(A) P–(ii), Q–(iii), R–(i), S–(iv)

(B) P–(ii), Q–(i), R–(iii), S–(iv)

(C) P–(i), Q–(iii), R–(ii), S–(iv)

(D) P–(iv), Q–(iii), R–(i), S–(ii)

27. A river that flows through multiple states is heavily used for irrigation, drinking, and industrial purposes. During summer, the river dries up in many areas, and water pollution from untreated sewage and industrial effluents has made the

water unfit for use. Conflicts arise between upstream and downstream states over water sharing.

Which combination of actions would best ensure equitable and sustainable water use?

(A) Upstream states should use water freely, and downstream states should adapt to scarcity.

(B) Implement inter-state water-sharing agreements, treat industrial and domestic wastewater, promote water-saving irrigation techniques, and involve communities in water management.

(C) Ban all agriculture in the river basin.

(D) Build high walls along the river to prevent downstream use.

28. Mixing acid X and base Y produces heat, solution $\text{pH} \approx 7$. Which is true?

(A) X = weak acid, Y = weak base

(B) X = strong acid, Y = strong base

(C) X = strong acid, Y = weak base

(D) X = weak acid, Y = strong base

29. A simple pendulum takes 32 seconds to complete 20 oscillations. If the length of this pendulum is increased to 4 times its original length, what will be the new time period?

(A) 1.6 seconds (B) 3.2 seconds

(C) 6.4 seconds (D) 0.8 seconds

30. During a blood donation camp, a person with Blood Group 'O' is often called a "Universal Donor." From a biological standpoint, why is this blood group safe to give to almost anyone?

(A) It contains all types of antibodies that fight every known disease.

(B) It lacks specific antigens on the surface of the Red Blood Cells, so the recipient's immune system doesn't "attack" the new blood.

(C) It has the highest concentration of hemoglobin, making it the most efficient for oxygen transport.

(D) It is the only blood group that does not contain any white blood cells.

31. A student observes that heating a piece of chocolate causes it to melt. Which of the following statements is true?

(A) The chocolate has undergone a chemical change because it cannot be solid again

(B) The chocolate has undergone a physical change; its composition remains the same

(C) Chocolate melting is an irreversible chemical change

(D) Melting chocolate produces a gas

32. Mr. Sharma drives his car from Delhi to Agra. At the start, the odometer reads 4500 km. After arriving in Agra 3 hours later, the odometer reads 4680 km. What is the average speed of the car in m/s?

(A) 60 m/s (B) 16.67 m/s

(C) 25 m/s (D) 50 m/s

33. A hilly region receives heavy rainfall in the monsoon but faces water shortages in winter. Many households rely on small streams and springs, but these often dry up outside the rainy season. The local government is considering building

small check dams, promoting community ponds, and planting trees to improve groundwater recharge.

Which reasoning best explains why these measures can solve the problem?

(A) Check dams and ponds store water, trees increase infiltration, and together they replenish groundwater, making water available year-round.

(B) Only planting trees is enough; check dams and ponds are unnecessary.

(C) Small check dams cannot store water, so building them wastes money.

(D) Water availability depends solely on rainfall, so human efforts are irrelevant.

34. A solution reacts vigorously with metals and carbonates. Its pH is:

- (A) 1–3 (B) 5–6
(C) 7 (D) 8–10

35. Tooth enamel is attacked by acids in soft drinks. Which base can neutralize the acid without damage?

- (A) NaOH (B) CaCO₃
(C) NH₄OH (D) HCl

36. A digital thermometer shows **104°F** for a patient. What is this temperature in **°C** and does it indicate fever? (Given: Normal body temperature $\approx$ **37°C**)

- (A) 40°C, yes fever
(B) 37°C, no fever
(C) 38°C, slight fever
(D) 42°C, very high fever

37. An athlete is training at sea level and then moves to a high-altitude training center. For the first two days, they feel extremely breathless. After two weeks,

they feel normal. What analytical change occurred in their blood?

(A) The White Blood Cell count increased to fight the cold air.

(B) The Red Blood Cell count increased, allowing the blood to carry more oxygen despite the "thin" air.

(C) Their heart stopped pumping blood to the skin to save oxygen for the muscles.

(D) The blood became completely transparent to allow light to stimulate the heart.

38. A fruit farmer wants to produce a new variety of apple that has the "sweetness" of Apple Type A and the "crunchiness" of Apple Type B. He uses a technique called "Grafting" (joining a stem of A to the root of B). Why does he use this instead of planting seeds from Type A?

(A) Seeds from Type A are too small to grow.

(B) Seeds are a result of sexual reproduction and may not have the exact same traits as the parent; grafting (vegetative propagation) ensures the exact traits are preserved.

(C) Grafting makes the fruit grow twice as fast.

(D) Apple trees are unable to produce seeds in modern orchards.

39. A farmer decides to grow a new field of sugarcane. Instead of buying seeds, he cuts pieces of the stem of an existing sugarcane plant and buries them in the soil. This method of reproduction is:

(A) Sexual reproduction, because a new plant is produced.

(B) Vegetative propagation, a form of asexual reproduction.

(C) Spore formation, because the stem contains microscopic spores.

(D) Fragmentation, because the stem was broken into pieces.

40. If a fuse wire is replaced by a **thick copper wire**, what can happen?

(A) Circuit becomes safer

(B) No current will flow

(C) Overheating and fire risk increases

(D) Bulb will glow dim

41. **Assertion (A):** Blood is a liquid tissue that consists of a fluid called plasma and various types of cells.

Reason (R): Plasma transports only oxygen, while Red Blood Cells transport food, Carbon dioxide, and nitrogenous wastes.

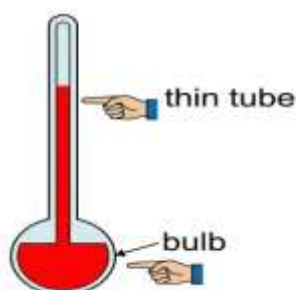
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

42. Look at this simple diagram of a thermometer:



The thin tube is made narrow mainly to:

(A) reduce heat transfer

(B) increase accuracy by making mercury rise noticeably

(C) prevent boiling of mercury

(D) make thermometer unbreakable

43. A light ray is incident on a plane mirror with an angle of incidence 40° . What is the angle of deviation?

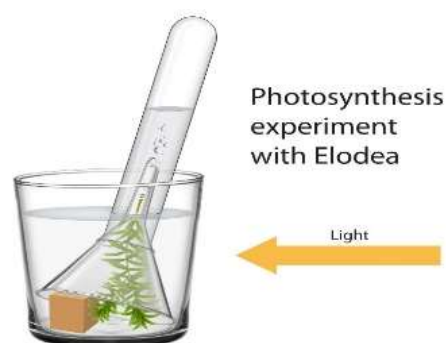
(A) 40°

(B) 80°

(C) 100°

(D) 140°

44. A student placed a sprig of the water plant Hydrilla in a funnel, inverted it in a beaker of water, and placed it in sunlight. They observed bubbles forming. If the student moves the beaker into a dark cupboard, what will happen to the bubble production?



(A) The bubbles will increase because plants breathe more in the dark.

(B) The bubbles will stop because the bubbles were oxygen produced during photosynthesis, which requires light.

(C) The bubbles will turn into starch grains and sink to the bottom.

(D) The bubbles will continue at the same rate but change from oxygen to nitrogen.

45. A current of 0.5 Amperes flows through an electric bulb for 10 minutes. **How much Electric Charge flows through the circuit during this time?**

(A) 5 Coulombs

(B) 300 Coulombs

(C) 50 Coulombs

(D) 3000 Coulombs

46. In an experiment to test for the necessity of sunlight for photosynthesis, a student covers a portion of a destarched leaf with black paper and leaves it in the sun. After 6 hours, they perform the Iodine test.

Which of the following results correctly identifies the "Control" and the "Variable" in this setup?

(A) Control: The covered part; Variable: The uncovered part.

(B) Control: The entire plant; Variable: The iodine solution.

(C) Control: The uncovered part (receiving light); Variable: The covered part (deprived of light).

(D) Control: The black paper; Variable: The starch.

47. A **student says:** "Regular reflection makes objects visible, but diffused reflection does not." Which is the best conclusion?

(A) Student is correct

(B) Student is wrong; diffused reflection mainly helps us see objects

(C) Both reflections never help in visibility

(D) Only mirrors help us see objects

48. An athlete runs on a circular track of radius 70 m. He completes exactly 1.5 rounds starting from point A. What is the ratio of the distance covered to the magnitude of his displacement? (Take $\pi = 22/7$)

(A) 11 : 7

(B) 22 : 7

(C) 33 : 7

(D) 3 : 2

49. A toy car requires 6 Volts to operate. You have standard dry cells of 1.5 V each. You connect them in series, but by mistake, you connect one cell with reversed polarity (Positive to Positive). If you used a total of 4 cells, what is the net voltage available to the car now?

(A) 6 V

(B) 4.5 V

(C) 3 V

(D) 0 V

50. An electromagnet is made using an iron nail, copper wire, and a cell. The student wants to make it stronger to lift more pins. Which combination will help MOST?

(A) Fewer turns of coil + weaker cell

(B) More turns of coil + high current

(C) More turns of coil + plastic nail

(D) Thicker nail only, without current

